



Mutual Fund Analytics Capstone Project

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DATA ANALYTICS

FINANCIAL RESEARCH

CAPSTONE PROJECT

Project Overview

Objective

Build an end-to-end analytics framework to analyze mutual fund performance, investor behavior, and category trends — enabling data-driven investment insights for fund houses and retail investors.

Business Problem

Mutual fund investors and advisors lack a unified platform to evaluate scheme performance, assess risk-adjusted returns, and identify optimal SIP strategies across fund categories and time horizons.



Dataset Overview

Five interconnected datasets form the analytical backbone of this project, covering the full spectrum of mutual fund operations.



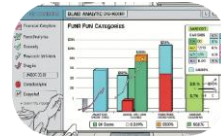
NAV History

Daily Net Asset Value records across schemes for return calculations.



SIP Inflows

Systematic Investment Plan inflow data by scheme, category, and period.



Category Inflows

Aggregate inflow trends across equity, debt, hybrid, and liquid categories.



Investor Transactions

Redemption, purchase, and switch transactions with investor demographics.



Scheme Performance

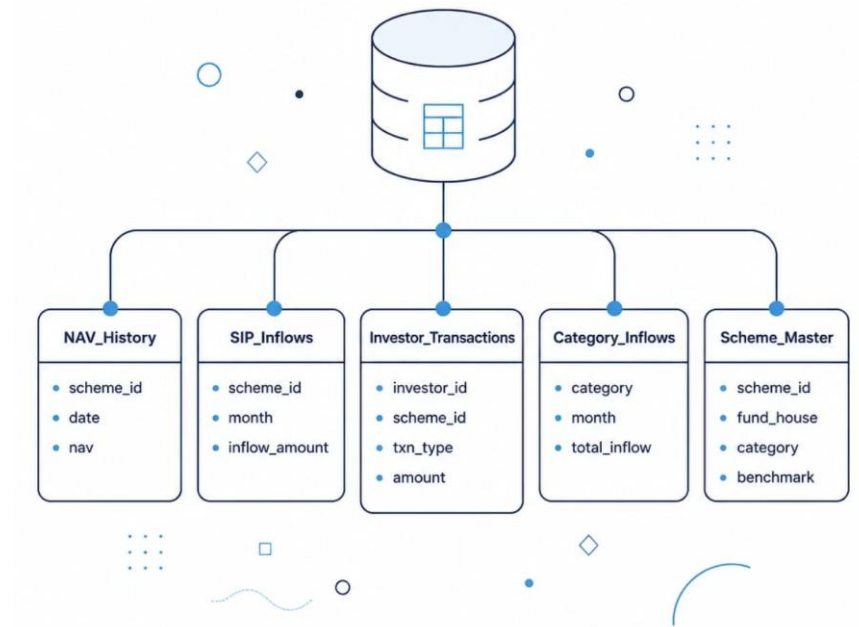
Benchmark comparisons, expense ratios, and AUM data per scheme.

ETL Pipeline



The pipeline ensures raw, heterogeneous fund data is standardized, validated, and loaded into a queryable SQLite database—forming a reliable foundation for downstream analysis and dashboarding.

Database Design



SQLite Architecture

A normalized relational schema was designed to support complex analytical queries while maintaining data integrity across millions of transaction records.

5 Core Tables

Linked via foreign keys for seamless joins

Optimized Queries

CTEs and window functions for advanced analytics

Primary and foreign key relationships ensure data consistency across all analytical modules.

Exploratory Data Analysis

Top Fund Houses

SBI, HDFC, and ICICI Prudential dominate AUM share, collectively holding over 45% of total industry assets under management.

SIP Trends

Monthly SIP inflows grew consistently, with equity categories witnessing the steepest adoption curve among retail investors.

Investor Demographics

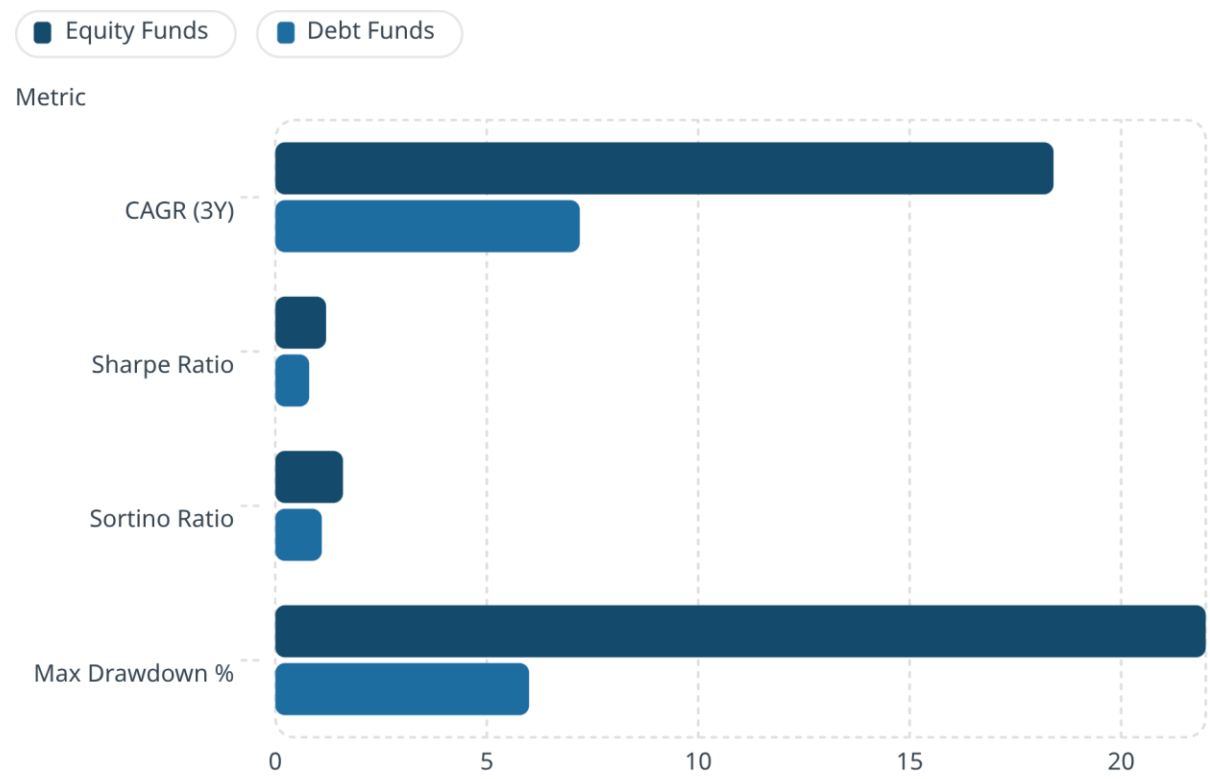
Age group 25–40 accounts for the highest SIP participation, reflecting growing financial literacy among millennials.

Category Inflows

Equity funds led net inflows, while debt categories showed resilience during market volatility periods.



Performance Analytics



Risk-Adjusted Metrics

A comprehensive suite of performance metrics was computed to evaluate fund efficiency beyond raw returns.

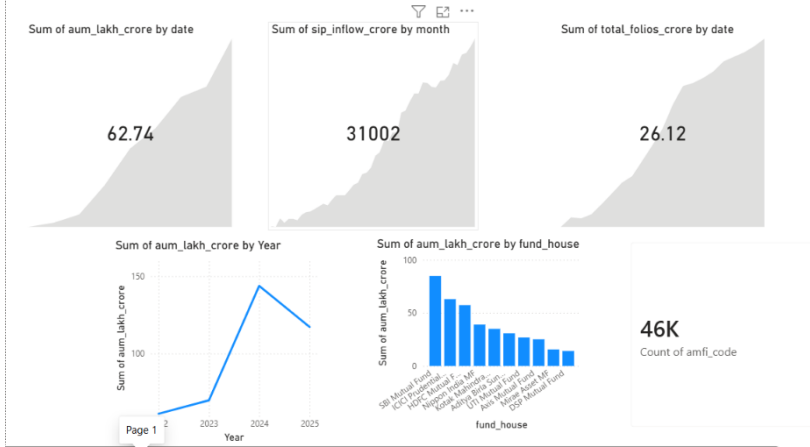
- **CAGR** — Compounded annual growth over 1, 3, and 5-year windows
- **Sharpe & Sortino** — Risk-adjusted return per unit of total and downside deviation
- **Alpha & Beta** — Excess return and market sensitivity vs. benchmark
- **Tracking Error** — Deviation from benchmark performance
- **Maximum Drawdown** — Peak-to-trough decline measuring worst-case loss

Key Results

- Highest Sharpe Ratio Fund: **Kotak Emerging Equity Fund** → 0.96
- Top High-Risk Recommendation: **ICICI Prudential Midcap Fund**
- Risk Analysis: **Equity funds showed higher downside risk in VaR analysis.**
- Drawdown Observation: **Large-cap funds experienced relatively lower drawdowns.**

Interactive Dashboard

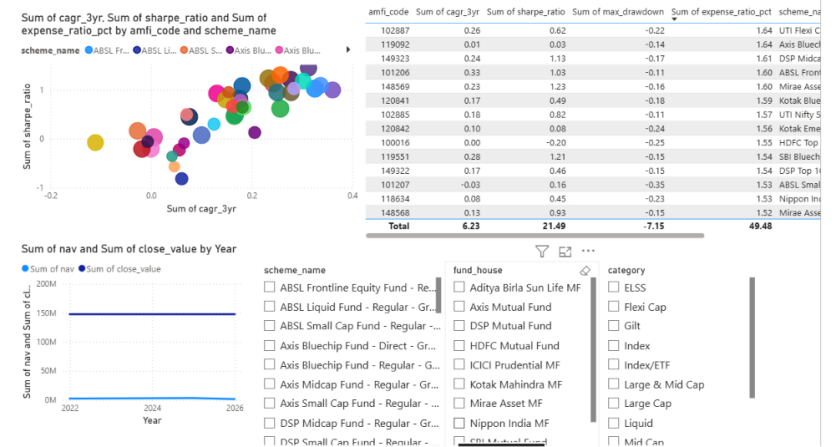
Mutual Fund Industry Overview



KPI Cards

→ Total AUM, SIP count, CAGR, and Sharpe Ratio at a glance

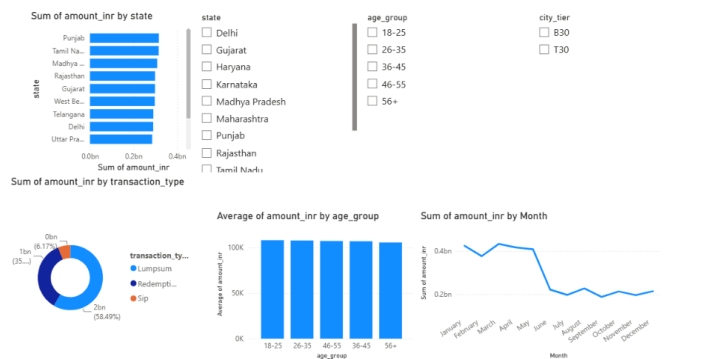
Fund Performance Dashboard



Fund Analysis

→ Drill-down views by fund house, category, and scheme

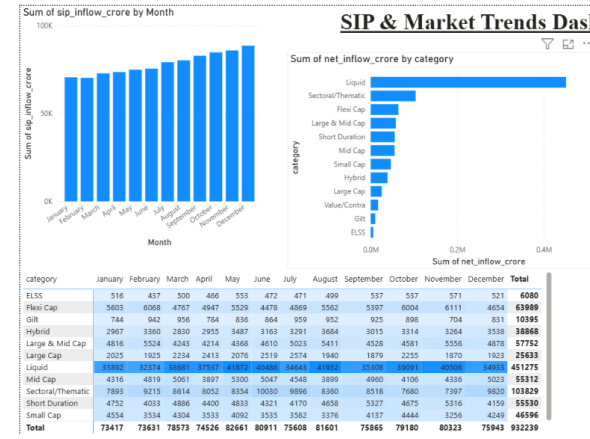
Investor Analytics Dashboard



Investor Analysis

→ Demographic breakdowns and transaction patterns

SIP & Market Trends Dashboard



→ Slicers & Filters

Interactive controls for time period, category, and risk profile

Advanced Analytics



VaR & CVaR

Value-at-Risk and Conditional VaR computed at 95% and 99% confidence levels to quantify tail risk exposure across fund categories.



Rolling Sharpe Ratio

12-month rolling Sharpe ratios reveal how risk-adjusted performance evolves over time, highlighting consistency and volatility clusters.



Cohort Analysis

Investor cohorts segmented by entry period show SIP retention rates and redemption behavior over 6, 12, and 24-month horizons.



SIP Continuity Analysis

Measures the proportion of investors maintaining uninterrupted SIPs — a key indicator of financial discipline and fund loyalty.



Fund Recommendation System

A rule-based engine ranks schemes by composite scores of CAGR, Sharpe, drawdown, and consistency for personalized investor profiles.

Key Insights

- 2024 investor cohort invested ₹349 Cr.
- Investors with average SIP gap >35 days were flagged as At-Risk.
- High-risk funds generated the highest risk-adjusted returns.
- Historical VaR and CVaR highlighted significant tail risk in equity schemes.

Key Insights & Conclusion

Top Performing Funds

Mid-cap and flexi-cap equity funds delivered the strongest risk-adjusted returns, outperforming large-cap peers on Sharpe and Sortino metrics.

Investor Behavior

SIP investors who stayed invested for 3+ years achieved significantly higher effective returns, validating the power of compounding and discipline.

Risk Insights

VaR analysis confirmed equity funds carry meaningful tail risk; CVaR highlighted the importance of downside protection in portfolio construction.

Future Improvements

Next steps include integrating ML-based return forecasting, expanding to international fund data, and deploying the dashboard on a cloud platform.



Thank You,

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GitHub Repository:

<https://github.com/zain-cmd-hub/mutual-fund-analytics.git>